Chapter 9. Virtual memory. Exercises

Original problems. State assumptions, units, and intermediate reasoning. Answers are in the separate solutions PDF.

OS09-E01 Reference-string replacement

Start with three empty frames and references 1,2,3,4,1,2,5,1,2,3,4,5; the four-frame comparison also starts empty. CLOCK has slots 0, 1, 2 and initially hand=0. Fill free slots in that order without moving the hand; a hit sets R=1 without moving it. When full, scan from the hand: clear R=1 and advance, then replace the first R=0, set the new page R=1, and advance the hand. OPT breaks equal next-use times, including never-used-again ties, by evicting the lower page number.

1. Compute FIFO, LRU, OPT, and CLOCK fault counts.
2. Give FIFO's victim sequence and identify hits that leave its queue unchanged.
3. Compare FIFO with three and four frames. Does increasing capacity increase faults?
4. Why does the anomaly not refute OPT's lower bound? What does exact LRU cost in hardware/metadata updates?

OS09-E02. Effective access time

A normal access costs 80 ns. A faulting reference costs 5 ms in total. Let fault probability be p and require EAT<=100 ns.

1. Derive the inequality.
2. Find the largest acceptable p.
3. Recompute when fault service rises to 10 ms under the same convention.

Check your approach. Convert units first and state whether the fault term includes completion of the reference.

OS09-E03. Working sets and thrashing

There are 64 usable frames. Three processes touched 20,24,28 different pages in their respective recent windows.

1. Compare the simple sum with available frames.
2. Does that prove current thrashing?
3. Which control variables can the system change?

Check your approach. A working-set estimate predicts pressure; runtime evidence establishes its consequences.

OS09-E04 Demand and COW faults

4 KiB pages,3 full frames:F0→clean P0, F1→dirty P1, F2→clean P2. P5 is a legal readable nonresident page; choose F1. Read and writeback each take 8 ms without overlap; ignore other overhead. Assume page locking succeeds, no concurrent modification occurs, and all affected CPUs acknowledge required shootdown before frame reuse. In an independent COW experiment F23 has already been safely obtained; parent/child share F17(value 41, refs 2, COW, read-only), and child stores 42.

1. Trace P5 read, I/O counts, TLB events, and service time.
2. What changes with clean P2 as victim?
3. Why is the COW write not a missing-page fault? Show both PTEs/F17/F23.
4. Give two handler defects causing repeated faults or corruption.

OS09-E05 Quotas and local/global victims

Part 1 is an independent 64-frame quota model with sizes A100/B200/C300. Parts 2–4 use a separate full six-frame model:A={1,2,3}, B={4,5}, C={6}; last uses A1=10, A2=5, A3=8, B4=2, B5=7, C6=9. A faults 7, no free frames.

1. Floor proportional shares and assign the remainder by largest fraction.
2. Find local/global victims and final resident sets.
3. Which spreads pressure to B? Give the immediate next fault.
4. Can fairer allocation fix residency below a correctness minimum?

OS09-E06 Working sets and PFF

Window Δ=4. A references 1,2,3,1,2,4,5,1; B:6,7,6,8,7; C:9,9,9. Frames 6. Assume the referenced pages are not shared between processes, so their sum counts distinct backing pages. PFF upper 0.4/lower 0.1; three five-event windows are FFFFF, FHHHH, HHHHH.

1. Give all working sets, WSS, D, and meaning of D>m.
2. Compute rates/actions and explain two thresholds.
3. What if high PFF has no free frame?
4. GiveD>m without actual thrashing and identify runtime evidence needed.

OS09-E07 Capacity monotonicity and offline optimality

Initially empty residents, an arbitrary finite reference string, unit fault cost, no prefetch benefit. Compare exact LRU/OPT.

1. Prove LRU resident inclusion using a capacity-independent recency stack, then nonincreasing faults.
2. Why cannot FIFO/CLOCK inherit that proof?
3. Prove OPT's farthest-next-use choice by first-divergence exchange. Which premise fails with dirty I/O costs?

OS09-E08 COW and stale-TLB isolation

F1 has a writable P1 mapping still cached by CPU1. In a separate scenario, P1/P2 MAP_PRIVATE-map the same file-backed 4 KiB page F40; both PTEs are read-only and managed as COW, with page and source-file value 7 and no external or concurrent writes. For part 3 independently assume that after child COW the parent points to a private anonymous page with one remaining user mapping.

1. CPU0 clears the PTE and loads P5 into F1 before CPU1 stores through P1. What happens, and what order is required?
2. P2 alone writes 9, then P1 writes 8. Give mappings, values, copy bytes, and source-file content.
3. After child COW leaves one parent mapping, must parent PTE change immediately? When can a later fault upgrade without copying?

OS09-E09 Conditional Linux 3.2 reclaim

Use asynchronous order-0, kswapd=true, priority 9, may_writepage/FS/I/O/unmap allowed, page locks succeed, with no concurrent re-reference or modification, and P1 is not under lumpy reclaim. P1 is PTE-referenced anonymous; P2 is a clean file page with no extra pin and removable mapping/references; P3 is a dirty file page with PTE reference=0, old PageReferenced=0, and a successfully submitted asynchronous pageout; P4 is an already isolated mlocked page before its cull to unevictable; P5 is under writeback. P4 is a transient isolated candidate, not an assertion that mlocked pages normally remain on the ordinary inactive LRU.

1. Give every first-round action and free increment.
2. Change P3 to direct reclaim or priority 10. Which branch changes?
3. Change P1 to file, first PTE reference, old PageReferenced 0. What happens?
4. Replace P4's mlock with an extra DMA pin on ordinary LRU. Are they equivalent? Does completed P3 writeback alone allow reclaim?

OS09-E10 Can a window length replace an LRU capacity?

Use references 1,2,1,3,1,2,4,1, with working-set window Δ=4 and initially empty exact LRU. Pages are equal-sized; there is no prefetching or pinning.

(1) Give the window, WS, and WSS at t=4 through 8. Why does expiration of a page-1 reference at t=5 not remove page 1 from WS?
(2) Give each pre-access rank r, both F/H traces at capacities 2 and 3, and fault totals.
(3) At t=7 compare WS with three-frame LRU residents. Derive the hit condition using d and explain its off-by-one relation to r.

References and attribution

Lin Cing, handwritten operating systems notes. Silberschatz, Galvin and Gagne, Operating System Concepts, 9th ed., Chapter 9. Linux v3.2 vmscan.c; Linux v3.2 memory.c; mmap(2).

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https://github.com/xinhi6716/computer-science-self-study